String Questions

1) What is Strings?

2) How to define multi-line String Literals?

3) How many ways we can access characters of strings?

4) Write a Program to Accept some String from the Keyboard and display its Characters by Index wise (both Positive and Negative Index)

5) Given the string

```
text = "The quick brown fox jumps over the lazy dog and runs towards the forest."
```

```
What will be the output ?
```

```
a. text[0:3]
b. text[4]
c. text[-1]
d. text[::2]
e. text[-7:-1]
f. text[16:19]
g. text[10:5]
h. text[100:110]
i. text[5:10:-1]
j. text[4:4]
k. text[0:10:-1]
l. text[10:0:1]
m. text[20:10:2]
n. text[100:110:1]
o. text[-10:-1:-1]
p. text[-50:-20:-1]
q. text[-100:-80:-1]
r. text[-5:-5:-1]
s. text[-1:-10:-1]
t. text[-10:-1:1]
u. text[-10:-20:1]
v. text[-100:-80:1]
w. text[50:-50:1]
x. text[1:10:0]
y. text[10:12:5]
z. text[12:10:5]
aa.   text[-10:-30:2]
bb.   text[10:5:10]
cc.   text[0:70:10]
dd.   text[70:0:-10]
ee.   text[::6]
```

6) s = “python is easy language”
   Which function we can use to calculate the length of string and what is the length of above string?

7) What will be the output?
   s = “python is easy language”
   print(“python” in s) → ?
   print(“java” in s) → ?

8) Which method in strings used for removal of spaces.
   S = “ My Name is Python “

   a. Remove left & right spaces
   b. Remove left spaces
   c. Remove right spaces

9) How to find substrings inside main strings?
   S = “Python is easy language”
   a. print(s.find(‘easy’)) → ?
   b. Print(s.find(‘java’)) → ?
   c. print(s.index(‘easy’)) → ?
   d. Print(s.index(‘java’)) → ?
   e. Print(s.find(“easy”, 0, 15)) → ?

10) Write a Program to display all Positions of Substring in a given Main String?

   S = “abbababababacdefg”
   Sub = ‘a’

   Output should be :
   Found at position 0
   Found at position 3
   Found at position 5
   Found at position 7
   Found at position 9
   Found at position 11

11) S = “Python is Very Hard Language”

I want output as
S = “Python is Very Easy Language”

How to do ?

12) S = “aaaabbbbbaaaavvvffttrrezzaaaa”
I want to count, How many times “a” appeared, How to do ?

13) s = 'learning Python is very Easy'

using string method, Make this string as
a. lowercase
b. uppercase
c. title case
d. swap case
e. capitalize

14) S = “Python”
Reverse it with help of
a. Slicing operator
b. With help of logic

15) S= “Python Java Javascript Typescript”
Reverse above string.

Output = “Typescript Javascript Java Python”

16) S= “Python Java Javascript Typescript”
Reverse above string.

Output = “tpircsepyT tpircsavaJ avaJ nohtyP”

17) Write a Program to Merge Characters of 2 Strings into a Single String by taking Characters alternatively.

S1 = “Python”

S2 = “Java”

Output should be “PJyatvhaon”

18) Write a Program to Sort the Characters of the String and First Alphabet Symbols followed by Numeric Values.

Input: B4A1D3

Output: ABD134

19) Write a Program for the following Requirement

Input: “a4b3c2”

Output: “aaaabbbcc”

20) Write a Program for the following Requirement

Input: “abc”

Output: “aababc”

21) Write a Program to perform the following Activity

Input: “a4k3b2”

Outpt: “aeknbd”

22) Write a Program to Remove Duplicate Characters from the given Input String?

Input: “ABCDABBCDABBBCCCDDEEEF”

Output: “ABCDEF”

23) Write a program to Check Whether the String is Symmetrical or Palindrome.

Input = “amaama”

24) Write a program Python Program For Converting Roman Numerals To Decimal?

Answers

Q4 ->

```
s = input("Enter Some String: ")
i = 0
for x in s:
    print(f"The character present at positive index {i} and at negative index {i - len(s)} is {x}")
    i += 1
```

Q10 ->

```
s=input("Enter main string: ")
subs=input("Enter sub string: ")
flag=False
pos=-1
n=len(s)
while True:
    pos=s.find(subs,pos+1,n)
    if pos==-1:
        break
    print("Found at position",pos)
    flag=True
if flag==False:
    print("Not Found")
```

Q15

main.py

```python
s = “Python Java Javascript Typescript”
print(type(s))
s1 = s.split()
print(s1)
print(type(s1))
s1.reverse()
print(s1)
s2 = " ".join(s1)
print(s2)
print(type(s2))
```

Output

```
<class 'str'>
['Python', 'Java', 'Javascript', 'Typescript']
<class 'list'>
['Typescript', 'Javascript', 'Java', 'Python']
Typescript Javascript Java Python
<class 'str'>

=== Code Execution Successful ===
```

Q16

main.py

```python
s = “Python Java Javascript Typescript”
print(type(s))
s1 = s.split()
print(s1)
s2 = []

for x in s1:
    output = ""
    i = len(x)-1
    while i>=0:
        output = output + x[i]
        i = i-1
    s2.append(output)

print(s2)
s2.reverse()
print(s2)
s3 = " ".join(s2)
print(s3)
```

Output

```
<class 'str'>
['Python', 'Java', 'Javascript', 'Typescript']
['nohtyP', 'avaJ', 'tpircsavaJ', 'tpircsepyT']
['tpircsepyT', 'tpircsavaJ', 'avaJ', 'nohtyP']
tpircsepyT tpircsavaJ avaJ nohtyP

=== Code Execution Successful ===
```

Q17

main.py

```python
s1=input("Enter First String: ")
s2=input("Enter Second String: ")
output=''
i = 0
j = 0
while i<len(s1) or j<len(s2):
    if i<len(s1):
        output=output+s1[i]
        i+=1
    if j<len(s2):
        output=output+s2[j]
        j+=1
print(output)
```

Output

```
Enter First String: Python
Enter Second String: Java
PJyatvhaon

=== Code Execution Successful ===
```

Q18

main.py

```python
s = input("Enter Some String: ")

s1 = "" # To store letters
s2 = "" # To store non-letters (numbers, symbols, spaces)

for x in s:
    if x.isalpha():
        s1 += x
    else:
        s2 += x

# Sorted() returns a list, so we join them back into strings
output = "".join(sorted(s1)) + "".join(sorted(s2))

print(output)
```

Output

```
Enter Some String: B4A1D3
ABD134

=== Code Execution Successful ===
```

Q19

main.py

```python
s = input("Enter Some String: ")
output = ''
previous = '' # Initialize to avoid errors

for x in s:
    if x.isalpha():
        output += x
        previous = x
    else:
        # Check if previous exists before trying to multiply
        if previous:
            output += previous * (int(x) - 1)

print(output)
```

Output

```
Enter Some String: a4b3c2
aaaabbbcc

=== Code Execution Successful ===
```

Q20

main.py

```python
s = input("Enter Some String: ") # Example: "abc"
output = ""

# Outer loop: determines how many characters to take (the 'stop' point)
for i in range(len(s)):

    # Inner loop: gathers characters from index 0 up to the current outer index
    for j in range(i + 1):
        output += s[j]

print(output)
```

Output

```
Enter Some String: abc
aababc

=== Code Execution Successful ===
```

Q21

main.py

```python
s = input("Enter Some String: ")
output = ''
previous = '' # Initialize to prevent NameError

for x in s:
    if x.isalpha():
        output += x
        previous = x
    else:
        # Check if we have a 'previous' character to shift
        if previous:
            # Shift the character by its Unicode value + the number
            new_char = chr(ord(previous) + int(x))
            output += new_char

print(output)
```

Output

```
Enter Some String: a4k3b2
aeknbd

=== Code Execution Successful ===
```

Q22

main.py

```python
s = input("Enter Some String: ")
l = []

for x in s:
    if x not in l:
        l.append(x)

output = ''.join(l)
print(output)
```

Output

```
Enter Some String: ABCDABBCDABBBCCCDDEEEF
ABCDEF

=== Code Execution Successful ===
```

Q23

main.py

```python
s = "amaama"

# --- 1. Palindrome Check ---
# Compare the string with its reverse
is_palindrome = s == s[::-1]

# --- 2. Symmetrical Check ---
# Split the string in half and compare the two halves
half = len(s) // 2
first_half = s[:half]
second_half = s[half:]
is_symmetrical = first_half == second_half

# Display Results
if is_palindrome:
    print(f'"{s}" is a Palindrome')
else:
    print(f'"{s}" is not a Palindrome')

if is_symmetrical:
    print(f'"{s}" is Symmetrical')
else:
    print(f'"{s}" is not Symmetrical')
```

Output

```
"amaama" is a Palindrome
"amaama" is Symmetrical

=== Code Execution Successful ===
```

Q24

[Python Program For Converting Roman Numerals To Decimal Lying Between 1 to 3999 - GeeksforGeeks]

Q5

text = "The quick brown fox jumps over the lazy dog and runs towards the forest."...

Based on your string `text = "The quick brown fox jumps over the lazy dog and runs towards the forest."`, here are the outputs for each slice:

Key	Slice	Output	Explanation
a.	`text[0:3]`	`'The'`	Indices 0, 1, 2.
b.	`text[4]`	`'q'`	Character at index 4.
c.	`text[-1]`	`'.'`	Last character.
d.	`text[::2]`	`'Teqikbonfxjmsoe h aydgadrn oad h oet'`	Every 2nd character.
e.	`text[-7:-1]`	`'forest'`	From -7 up to (not including) -1.

f.	`text[16:19]`	`'fox'`	Indices 16, 17, 18.
g.	`text[10:5]`	`''`	Forward slice, but start > stop.
h.	`text[100:110]`	`''`	Indices are out of bounds.
i.	`text[5:10:-1]`	`''`	Backward step, but start < stop.
j.	`text[4:4]`	`''`	Start and stop are equal.
k.	`text[0:10:-1]`	`''`	Backward step, but start < stop.
l.	`text[10:0:1]`	`''`	Forward step, but start > stop.
m.	`text[20:10:2]`	`''`	Forward step, but start > stop.
n.	`text[100:110:1]`	`''`	Out of bounds.
o.	`text[-10:-1:-1]`	`''`	Backward step, but -10 is left of -1.
p.	`text[-50:-20:-1]`	`''`	Backward step, but -50 is left of -20.
q.	`text[-100:-80:-1]`	`''`	Out of bounds (negative).
r.	`text[-5:-5:-1]`	`''`	Start and stop are equal.
s.	`text[-1:-10:-1]`	`'.tserof e'`	From end, moving 9 chars backward.
t.	`text[-10:-1:1]`	`'the fores'`	From -10 moving forward to -1.

u.	`text[-10:-20:1]`	`''`	Forward step, but -10 is right of -20.
v.	`text[-100:-80:1]`	`''`	Out of bounds (negative).
w.	`text[50:-50:1]`	`''`	Forward step, but 50 is right of -50.
x.	`text[1:10:0]`	**ValueError**	Slice step cannot be zero.
y.	`text[10:12:5]`	`'u'`	Char at 10; next (15) is past stop (12).
z.	`text[12:10:5]`	`''`	Forward step, but start > stop.
aa.	`text[-10:-30:2]`	`''`	Forward step, but -10 is right of -30.

bb.	`text[10:5:10]`	`''`	Forward step, but start > stop.
cc.	`text[0:70:10]`	`'Tir n h d'`	Every 10th character.
dd.	`text[70:0:-10]`	`' r h n ri'`	Every 10th character backward.
ee.	`text[::6]`	`'T kofm rldnus t'`	Every 6th character.